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AI IN HEALTHCARE
SYMPOSIUM

Improving Pre-operative Assessment Quality with Dual-Agent AI Health Questionnaires

— a prototype and study proposal

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INTRODUCTION

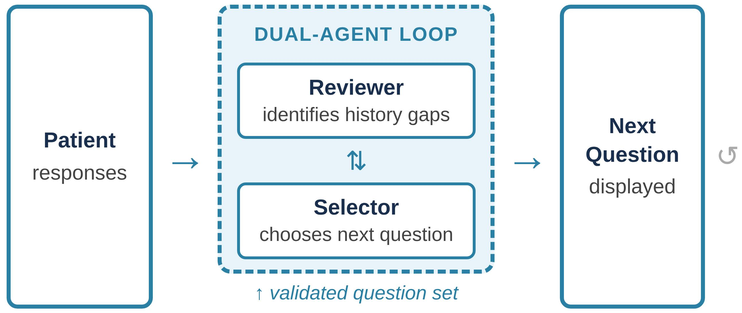
Accurate pre-operative patient history is essential for surgical safety. Standard questionnaires face a fundamental tradeoff — longer forms are more comprehensive but yield lower completion rates and accuracy. Static forms cannot adapt to individual patient profiles, leading to irrelevant questions and missed clinically important information.

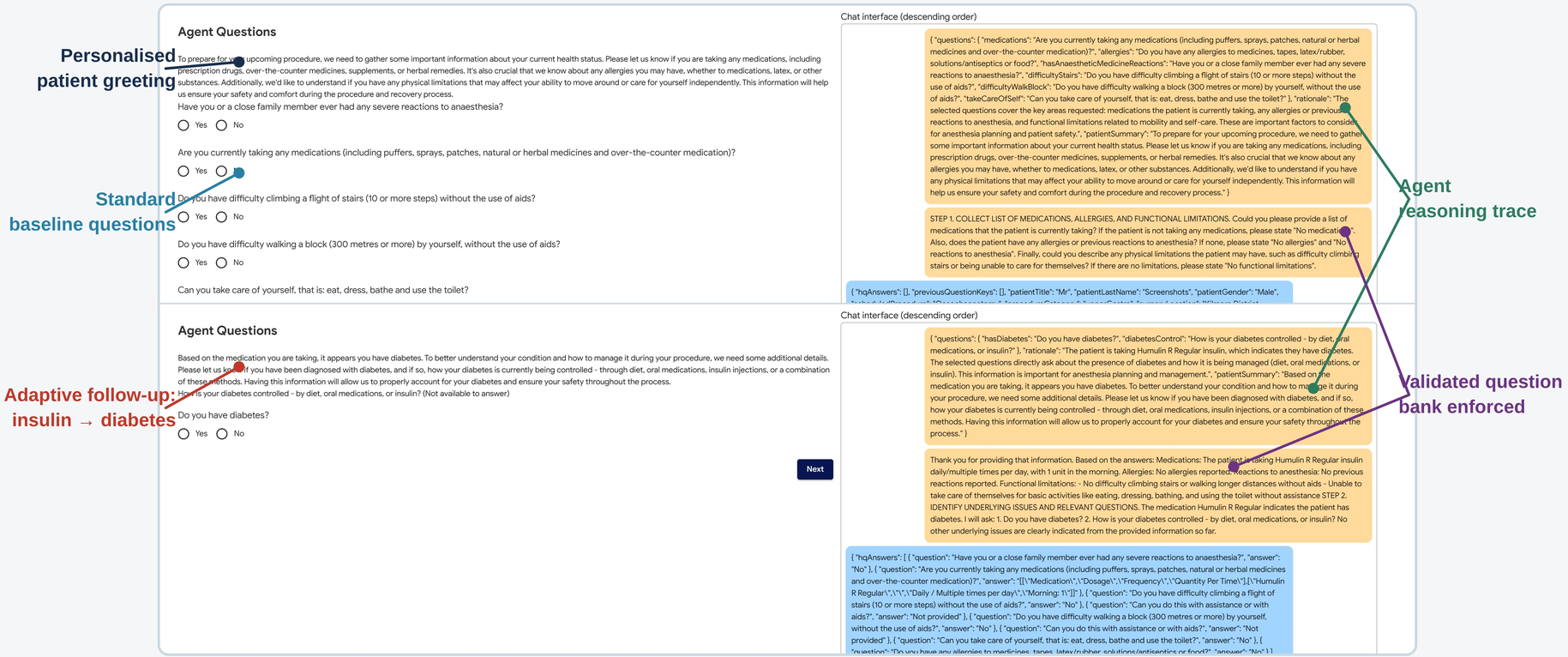
AIMS

To obtain higher quality pre-operative patient history by asking targeted, patient-specific questions based on each patient's profile and prior responses — naturally reducing questionnaire length as a consequence of improved relevance.

METHODS

A dual-agent LLM system dynamically tailors questionnaires to each patient. The system iterates until the history is sufficiently complete. It is model-agnostic, with LLM components interchangeable across foundation models.





RESULTS

A prototype was tested on pre-operative scenarios. In one example, a patient's insulin use triggered targeted follow-up questions on diabetes management and hypoglycaemia — questions a static form would have missed. An experienced clinician validated the clinical relevance of the questions asked.

CONCLUSIONS

This proof-of-concept demonstrates that a dual-agent GenAI system can dynamically tailor pre-operative health questionnaires, asking clinically relevant targeted questions while suppressing irrelevant ones. This suggests potential to improve both history completeness and patient experience.

FUTURE WORK

- Prospective clinical trial to measure history quality vs. standard questionnaires
- Multi-language support evaluation
- Integration with hospital pre-admission and EMR systems